

IMAGE STEGANOGRAPHY TOOL

Hide Text and Files Inside Images

ABSTRACT

This project implements a sophisticated steganography tool that enables users to securely hide confidential text messages or complete files within digital images using the Least Significant Bit (LSB) technique. The application features an intuitive graphical user interface built with Tkinter and supports drag-and-drop functionality for seamless user interaction. The tool supports PNG and BMP image formats, ensuring lossless data embedding and extraction. This implementation demonstrates practical applications of information security concepts including data concealment, secure communication, and digital watermarking.

INTRODUCTION

Steganography is the practice of concealing information within other non-secret data to avoid detection. Unlike cryptography which scrambles data, steganography hides the very existence of the message. This project focuses on image steganography using the LSB substitution method, where secret data is embedded by modifying the least significant bits of pixel color values. This technique is imperceptible to human eyes as the changes are minimal. The developed tool provides both embedding and extraction capabilities, allowing users to hide sensitive information in plain sight. Applications include secure communications, copyright protection, and covert data transmission in scenarios where encrypted messages might raise suspicion.

TOOLS AND TECHNOLOGIES

Tool/Library	Version	Purpose
Python	3.x	Primary programming language
Pillow (PIL)	10.x	Image processing and manipulation
Tkinter	Built-in	Graphical user interface framework
TkinterDnD2	0.3.0	Drag-and-drop functionality

IMPLEMENTATION METHODOLOGY

1. LSB Embedding Algorithm: The cover image is loaded and converted to RGB format. The secret message is converted to binary representation (8 bits per character). For each pixel's RGB values, the least significant bit is replaced with one bit from the secret message. A delimiter (<<<END>>>) is appended to mark the end of the hidden data. The modified pixels are saved as a new stego image in lossless PNG or BMP format.

2. Data Extraction Process: The stego image is loaded and pixel data is read sequentially. The LSB of each RGB component is extracted to reconstruct the binary message. The binary data is converted back to text characters. The extraction stops when the delimiter is detected. If the data represents a file (detected by FILE: prefix), it is decoded from hexadecimal and saved to disk.

3. GUI Development: A tabbed interface was designed with separate panels for embedding and extraction operations. Drag-and-drop support enables users to simply drop images onto designated areas. Real-time validation ensures image formats are compatible and message size doesn't exceed image capacity. Interactive file browsers allow selecting cover images, files to hide, and save locations for output.

4. File Hiding Capability: Binary files are read and converted to hexadecimal representation. The filename and hex data are packaged with a special prefix. During extraction, the file is reconstructed from hex data and saved with its original name, enabling complete file recovery.

KEY FEATURES

- Dual-mode operation supporting both text messages and complete file embedding
- LSB steganography ensuring imperceptible image modifications
- Support for PNG and BMP formats with lossless compression
- Drag-and-drop interface for enhanced usability
- Capacity validation preventing data overflow
- Automatic delimiter-based message termination
- File type preservation during extraction

CONCLUSION

This steganography tool successfully demonstrates practical implementation of information hiding techniques. The LSB method provides a balance between security and usability, making it suitable for everyday confidential communications. The intuitive GUI removes technical barriers, allowing non-experts to utilize steganography effectively. Future enhancements could include password encryption of hidden data, support for additional image formats (JPEG with special handling), and advanced algorithms like PVD (Pixel Value Differencing) for improved security. The project highlights the importance of steganography in modern cybersecurity, complementing traditional encryption methods by adding an additional layer of obscurity. By hiding information in plain sight, this tool provides a valuable resource for secure communication in sensitive scenarios.